MODULE 28 · DOMAIN 4 - CLOUD NATIVE ARCHITECTURE

Community and collaboration

The CNCF, project maturity, how Kubernetes is governed and released, open standards, and the roles people play.

DOMAIN 4 of 4 EXAM WEIGHT 12% MODULE 28 of 35

admin@sthithapragna.com

Kubernetes is not one company's product. It is **open source, governed in the open**, by a community.

The Cloud Native Computing Foundation

- **Founded in 2015** under the Linux Foundation, with Kubernetes as its first project.
- **A vendor-neutral home** for cloud native open-source projects.
- **Runs KubeCon + CloudNativeCon,** the certifications, and the CNCF Landscape.
- Google donated Kubernetes; no single company controls it now.

> ON THE EXAM
>
> The CNCF is part of the Linux Foundation. The OCI is a separate Linux Foundation project (module 2).

Who does what inside the CNCF

BODY	RESPONSIBLE FOR
Governing Board	Budget, marketing and business decisions
Technical Oversight Committee (TOC)	Technical vision; admits projects; approves moves between maturity levels
Technical Advisory Groups (TAGs)	Expertise in areas such as security, to support the TOC and projects
End User community	Organisations that use cloud native, feeding back their needs

ON THE EXAM

Technical decisions about projects, including graduation, belong to the TOC, not the Governing Board.

Project maturity levels

LEVEL	MEANS	TYPICAL ADOPTERS
Sandbox	Early, experimental, low barrier to entry	Innovators
Incubating	Growing production use, healthy contributor base	Early adopters
Graduated	Widely adopted, diverse maintainers, security audit, mature governance	Early majority
Archived	No longer active; kept for reference	Should migrate away

ON THE EXAM

Sandbox, incubating, graduated. The TOC decides each move.

A project's journey

01
Apply
The project applies to join the CNCF.

›

02
Sandbox
Accepted by the TOC as an early-stage project.

›

03
Incubating
Due diligence: adopters, contributors, governance.

›

04
Graduated
TOC vote, security audit, broad adoption.

›

05
Archived
Possible at any stage, if a project winds down.

ON THE EXAM

Graduation requires, among other things, a third-party security audit and committers from several organisations.

Projects from this course, and their level

PROJECT	LEVEL	MODULE
Kubernetes, etcd, containerd, CoreDNS	Graduated	3, 5, 2, 14
Prometheus, Fluentd, Jaeger, OpenTelemetry	Graduated	26
Envoy, Istio, Linkerd, Cilium	Graduated	13, 16
Helm, Argo, Flux, Knative, KEDA	Graduated	23, 24, 27
Falco, Kyverno, cert-manager, Harbor	Graduated	18, 19
SMI, OpenTracing	Archived	16, 26

ON THE EXAM

Check cncf.io/projects before the exam: levels change several times a year.

Currency check: recent moves

- **Graduated:** Knative, October 2025; Dragonfly, January 2026; Kyverno, March 2026.
- **Graduated:** OpenTelemetry, May 2026; Cloud Native Buildpacks, August 2026.
- **Archived:** SMI in 2023, replaced by GAMMA; OpenTracing, replaced by OpenTelemetry.
- Question banks that call these incubating, or current, are simply dated.

> ON THE EXAM
>
> If an answer depends on a maturity level, trust the most recent source you have seen.

Kubernetes work is organised by special interest groups, and changed through enhancement proposals.

admin@sthithapragna.com

How the Kubernetes project is governed

GROUP	ROLE
Steering Committee	Overall governance of the Kubernetes project
Special interest groups (SIGs)	Long-lived; each owns an area of the code and its decisions
Working groups	Temporary, cross-SIG, focused on one goal
Committees	Specific duties: Code of Conduct, Security Response

ON THE EXAM

SIGs own code. Working groups own a temporary goal. The Steering Committee oversees the whole.

SIGs, and where you met their work

SIG	OWNS	MODULE
SIG Node	The kubelet, runtimes, Pods	2, 6
SIG Network	Services, DNS, kube-proxy, Gateway API	13, 14, 15
SIG Storage	Volumes, PVs, CSI	20
SIG Auth	Authentication, RBAC, ServiceAccounts	17
SIG Scheduling	The scheduler	12
SIG Apps	Workload controllers	7, 8
SIG Cluster Lifecycle	kubeadm, Cluster API	11
SIG Autoscaling	HPA, VPA, Cluster Autoscaler, Karpenter	27

ON THE EXAM

A feature's owner is the SIG whose area it touches: networking questions belong to SIG Network.

How a feature becomes stable: the KEP process

01
Discuss
The idea is raised with the owning SIG.

02
Write a KEP
Motivation, design, risks, graduation criteria.

03
Approve
The SIG reviews and approves it.

04
Alpha
Behind a feature gate, off by default.

05
Beta, then GA
Proven in use, then stable. Deprecation rules apply.

ON THE EXAM

Every alpha, beta and stable feature you met in this course arrived through a KEP.

The Kubernetes release cycle

- **About three minor releases a year,** each cycle roughly fifteen weeks.
- **A volunteer Release Team** runs each cycle, with shadows learning each role.
- **Key dates:** enhancements freeze, then code freeze, then release.
- **Patch releases** for the three maintained minor versions, roughly monthly.

> ON THE EXAM
>
> Features that miss the enhancements freeze wait for the next release. Nothing is rushed in.

Contributing

- **A contributor ladder:** member, reviewer, approver, subproject owner.
- **OWNERS files** in each folder list who can review and approve changes there.
- **Not only code:** documentation, issue triage, testing, release work, events.
- **Everything is open:** meetings, recordings, Slack, and a Code of Conduct for all.

> ON THE EXAM
>
> Anyone can contribute. Responsibility grows through the ladder, earned by contribution.

Open standards that tie the ecosystem together

STANDARD	COVERS	MODULE
OCI specifications	Images, runtimes, registries	2
CRI, CNI, CSI	Plugging in runtimes, networks, storage	2, 13, 20
Gateway API	Traffic into and between services	15, 16
OpenTelemetry, OTLP	Telemetry: traces, metrics, logs	26
CloudEvents	Describing events	27
SPIFFE	Workload identity	16

ON THE EXAM

Standardise the interface, compete on the implementation.

Why open standards matter

- **No lock-in:** swap containerd for CRI-O, or Calico for Cilium, without touching workloads.
- **Interoperability:** any OCI image runs on any compliant runtime.
- **Competition on quality,** not on proprietary interfaces.
- **Portability:** conformant Kubernetes behaves the same on every certified distribution.

> ON THE EXAM
>
> 'Change the runtime or network plugin without changing workloads' is made possible by CRI and CNI.

Roles in a cloud native organisation

ROLE	FOCUSES ON
Application developer	Writing the software; using the platform
Platform engineer	Building the internal platform: self-service, golden paths
Site reliability engineer (SRE)	Reliability, SLOs, error budgets, incident response
DevOps engineer	Bridging development and operations: pipelines, automation
Security engineer	Shifting security left, into every stage (DevSecOps)
Cluster operator	Running, upgrading and securing the clusters

ON THE EXAM

Platform engineers build the paved road; developers drive on it; SREs keep it safe.

Platform engineering

- **An internal developer platform** hides Kubernetes' complexity behind self-service.
- **Golden paths:** supported templates for a new service, with CI, GitOps and monitoring included.
- **Platform as a product:** developers are its customers; their feedback shapes it.
- **Backstage,** a CNCF project, is a common developer portal for platforms.

> ON THE EXAM
>
> Developers should not need to master every module of this course to ship a service.

The community, and where to learn

- **KubeCon + CloudNativeCon:** the CNCF's flagship conferences, on several continents.
- **Kubernetes Community Days (KCDs):** local, community-run events.
- **CNCF Ambassadors,** meetups, and the Kubernetes Slack.
- **Certifications:** KCNA, KCSA, CKA, CKAD, CKS. Holding all five makes you a Kubestronaut.

> ON THE EXAM
>
> KCNA is the entry point. KCSA is its security sibling; CKA, CKAD and CKS are hands-on.

Six community questions. For each, ask: **who decides**, **who owns**, or **which standard**?

> A CNCF incubating project has growing adoption and has completed a security audit.
> Its maintainers apply to become a graduated project.

Who approves the move to graduated?

A The CNCF Governing Board

B The CNCF Technical Oversight Committee

C The Kubernetes Steering Committee

D The project's own maintainers

◆ PAUSE AND DECIDE

Who approves the move to graduated?

A The CNCF Governing Board

Handles budget and business, not technical maturity.

B The CNCF Technical Oversight Committee

The TOC admits projects and approves every level change.

C The Kubernetes Steering Committee

Governs Kubernetes only, not other CNCF projects.

D The project's own maintainers

They apply; they cannot approve their own graduation.

THE SIGNAL

Project maturity is a technical decision: the TOC.

> A new open-source tool, built by one company, has a handful of early users and an evolving design.
>
> The CNCF accepts it this month.

At which maturity level does it most likely enter?

A Graduated

B Incubating

C Sandbox

D Archived

◆ PAUSE AND DECIDE

At which maturity level does it most likely enter?

A **Graduated**
Needs broad adoption, diverse maintainers and an audit.

B **Incubating**
Needs real production adopters and due diligence.

C **Sandbox**
The entry point for early, experimental projects.

D **Archived**
For projects winding down, not joining.

THE SIGNAL

Early, experimental, one company, few users: sandbox.

> A contributor wants to add a significant new field to the Pod API, changing how the kubelet behaves.
>
> They want to follow the Kubernetes project's normal process.

What should they do first?

A Write a KEP and work through the owning SIG

B Open a pull request straight to the main branch

C Ask the CNCF TOC to approve the feature

D Email the Steering Committee for permission

◆ PAUSE AND DECIDE

What should they do first?

A Write a KEP and work through the owning SIG
Significant changes start with a KEP in the right SIG.

B Open a pull request straight to the main branch
Big API changes are not merged without a KEP.

C Ask the CNCF TOC to approve the feature
The TOC governs CNCF projects, not Kubernetes features.

D Email the Steering Committee for permission
The Steering Committee does not approve individual features.

THE SIGNAL

Significant Kubernetes change: a KEP, through the owning SIG.

> A proposed change touches Services, EndpointSlices and the Gateway API.
>
> The contributor needs to know which group owns this area of Kubernetes.

Which SIG owns it?

A SIG Node

B SIG Apps

C SIG Network

D SIG Storage

◆ PAUSE AND DECIDE

Which SIG owns it?

A **SIG Node**
Owns the kubelet, runtimes and Pods.

B **SIG Apps**
Owns workload controllers such as Deployments.

C **SIG Network**
Owns Services, DNS, kube-proxy and the Gateway API.

D **SIG Storage**
Owns volumes and CSI.

THE SIGNAL

Services and traffic belong to SIG Network.

> Quayside plans to replace containerd with CRI-O on its nodes, and Flannel with Cilium as its network plugin.
>
> No application manifest needs to change.

What makes this possible?

A Helm charts for every application

B GitOps with Argo CD

C A service mesh in every namespace

D The CRI and CNI open standards

◆ PAUSE AND DECIDE

What makes this possible?

A Helm charts for every application

Packages apps; says nothing about runtimes or networks.

B GitOps with Argo CD

Deploys apps; does not decouple the runtime.

C A service mesh in every namespace

Adds a traffic layer; unrelated to swapping plugins.

D The CRI and CNI open standards

Kubernetes talks to runtimes and plugins through standard interfaces.

THE SIGNAL

Swap a runtime or network plugin freely: CRI and CNI.

> At Quayside, one team builds self-service templates, so that other teams get a new service with CI, GitOps and monitoring already wired in.
>
> Application teams can ship without learning most Kubernetes internals.

Which role best describes that team?

A Site reliability engineers

B Platform engineers

C Application developers

D FinOps practitioners

◆ PAUSE AND DECIDE

Which role best describes that team?

A Site reliability engineers

Focus on reliability, SLOs and incidents.

B Platform engineers

Build the internal platform and its golden paths.

C Application developers

The customers of the platform, not its builders.

D FinOps practitioners

Focus on cost visibility and optimisation.

THE SIGNAL

Self-service templates and golden paths: platform engineering.

Module 28 in five lines

- The **CNCF** (Linux Foundation, 2015) hosts projects; the **TOC** decides maturity.
- **Sandbox, incubating, graduated**, or archived. Graduation needs a security audit.
- Kubernetes: **Steering Committee, SIGs, working groups**; changes via **KEPs**.
- About **three releases a year**; alpha behind feature gates, then beta, then GA.
- **Open standards** (OCI, CRI, CNI, CSI, Gateway API, OpenTelemetry) prevent lock-in.

That completes the teaching. Next: what changed, then confusion pairs, then practice exams.